Environmental Intelligence and Sustainability Analytics Platform
Week 6 – Interactive Visualization, Dashboard Development & Executive Decision Support

Team/Group 14

Executive Summary

Week 6 focused on transforming the analytical outputs developed during Week 5 into a comprehensive Business Intelligence solution using Power BI. The Environmental Intelligence & Sustainability Analytics Platform integrates climate, pollution, forest sustainability, forecasting, scenario analysis, and machine learning outputs into an executive-ready dashboard environment.

The completed solution includes six interactive dashboards covering environmental trends, climate change impact assessment, future scenario forecasting, environmental health monitoring, machine learning predictions, and executive-level sustainability reporting. Geographic visualizations, forecasting analytics, machine learning evaluation metrics, and executive KPIs were incorporated to provide decision-makers with actionable environmental intelligence.

The final platform supports sustainability monitoring, risk assessment, environmental forecasting, and strategic planning through an intuitive and interactive dashboard experience.

The completed visualization platform satisfies the Week 6 capstone deliverable by transforming analytical outputs into an interactive decision-support system for environmental monitoring and sustainability management.

1. Introduction

Week 6 represents the visualization and business intelligence phase of the Environmental Intelligence & Sustainability Analytics Platform.

Following the successful completion of data engineering, feature engineering, machine learning modeling, forecasting, clustering, and scenario analysis in Week 5, the objective of Week 6 was to transform analytical outputs into interactive dashboards that support executive decision-making.

Power BI was used to create an integrated environmental analytics platform capable of presenting historical trends, sustainability metrics, forecasting outputs, geographic insights, and machine learning results in a user-friendly format.

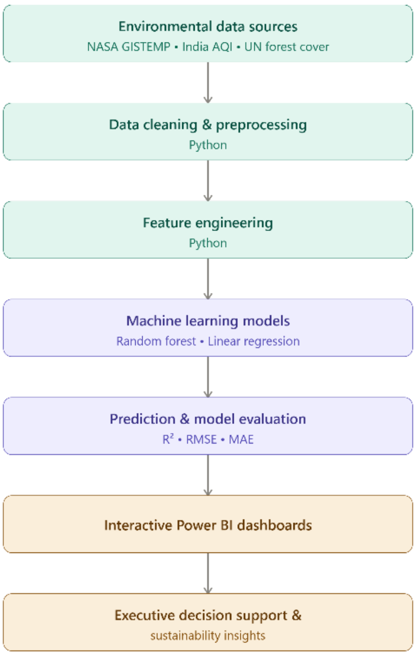
2. Week 6 Objectives

The key objectives for Week 6 were:

- Develop interactive Power BI dashboards
- Create executive-level KPI dashboards
- Build geographic and map visualizations
- Implement machine learning result visualization
- Present environmental forecasting results
- Support executive sustainability decision-making
- Create integrated environmental intelligence reporting
- Deliver an end-to-end business intelligence solution

3. Dashboard Architecture

Architecture Overview



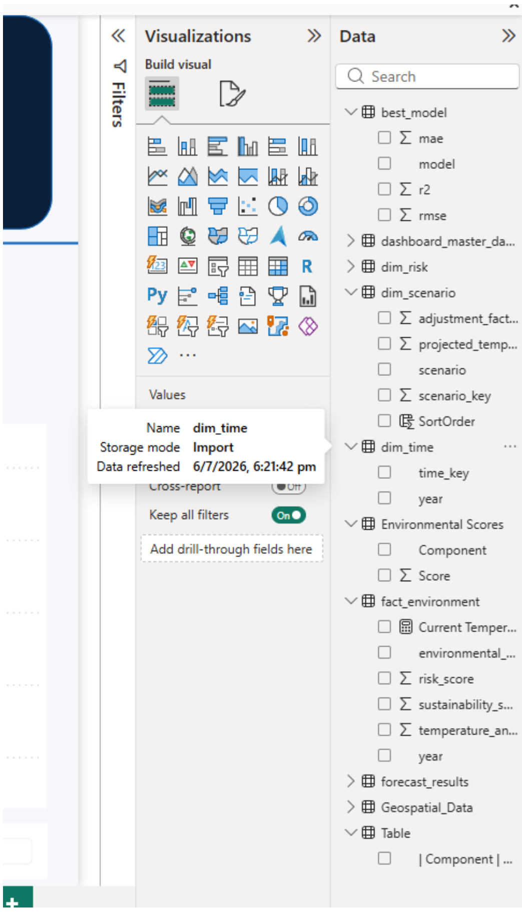
Description

The dashboard architecture follows a complete analytics lifecycle:

1. Environmental data was collected from NASA GISTEMP, India AQI, and UN Forest datasets.
2. Python was used for data cleaning and preprocessing.
3. Feature engineering generated environmental intelligence indicators.
4. Machine learning models (Random Forest and Linear Regression) were developed.
5. Forecasting and model evaluation were performed using R^2, RMSE, and MAE metrics.
6. Results were integrated into Power BI dashboards.
7. Executive decision support insights were generated for sustainability planning.

The architecture demonstrates a production-oriented environmental analytics pipeline capable of supporting forecasting, monitoring, and sustainability management.

4. Power BI Data Model



Data Model Description

A star-schema design was implemented to support analytical reporting and dashboard performance.

Fact Table

fact_environment

Contains:

- Environmental Health Index
- Risk Score
- Climate Score
- Pollution Score
- Sustainability Score
- Forecasting outputs

Dimension Tables

dim_time

Contains:

- Year
- Time Key

Used for trend analysis and forecasting.

dim_risk

Contains:

- Risk Categories
- Risk Drivers

Supports environmental risk analytics.

dim_scenario

Contains:

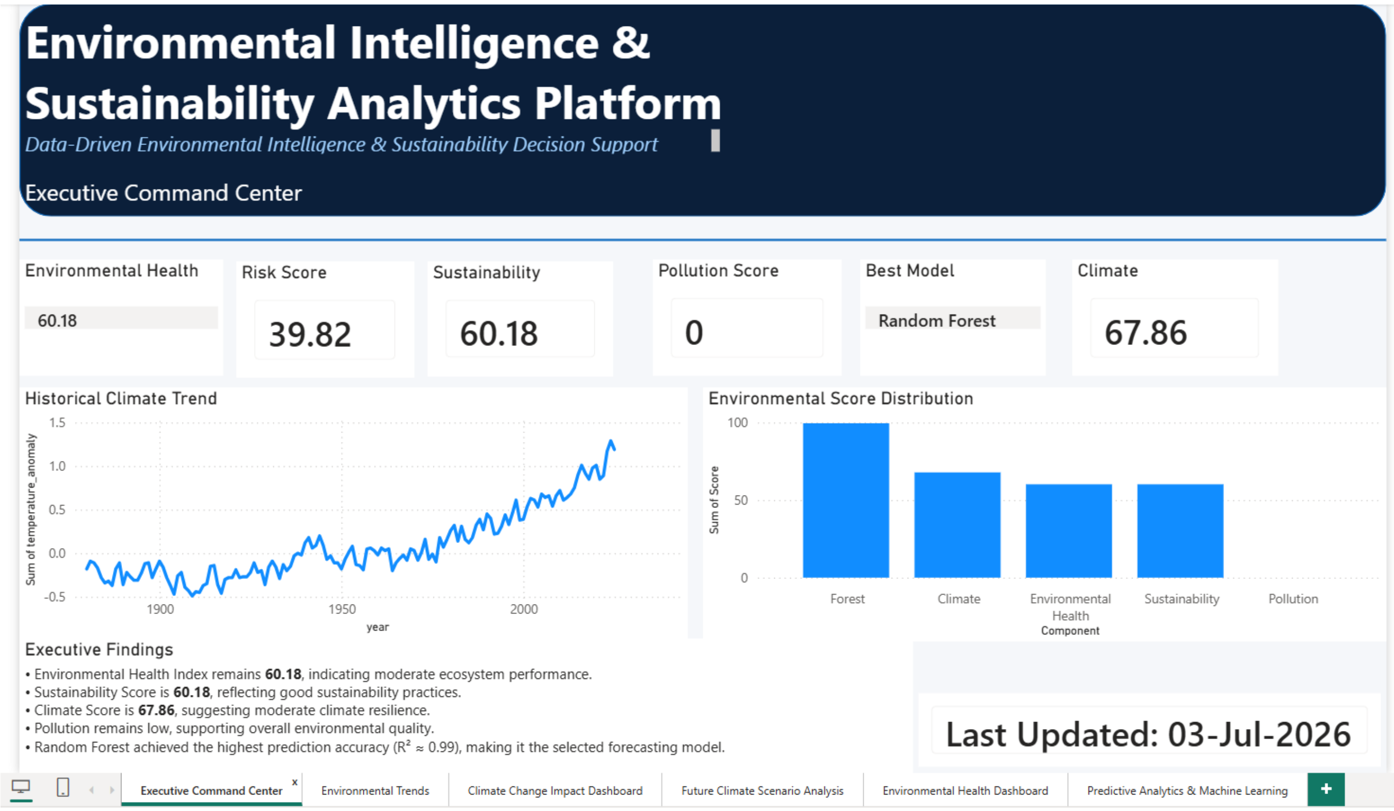
- Baseline Scenario
- Green Policy Scenario
- Aggressive Green Scenario
- Pollution Growth Scenario

Supports scenario forecasting analysis.

Relationships

The fact table is connected to dimension tables through unique surrogate keys, enabling efficient filtering, aggregation, and reporting within Power BI.

5. Executive Command Center



Overview

The Executive Command Center serves as the central decision-support dashboard.

Key KPIs

Environmental Health Index

60.18

Indicates moderate overall environmental performance.

Risk Score

39.82

Represents current environmental risk levels requiring monitoring.

Sustainability Score

60.18

Reflects sustainability performance across environmental dimensions.

Climate Score

67.86

Measures climate resilience and environmental stability.

Best Performing ML Model

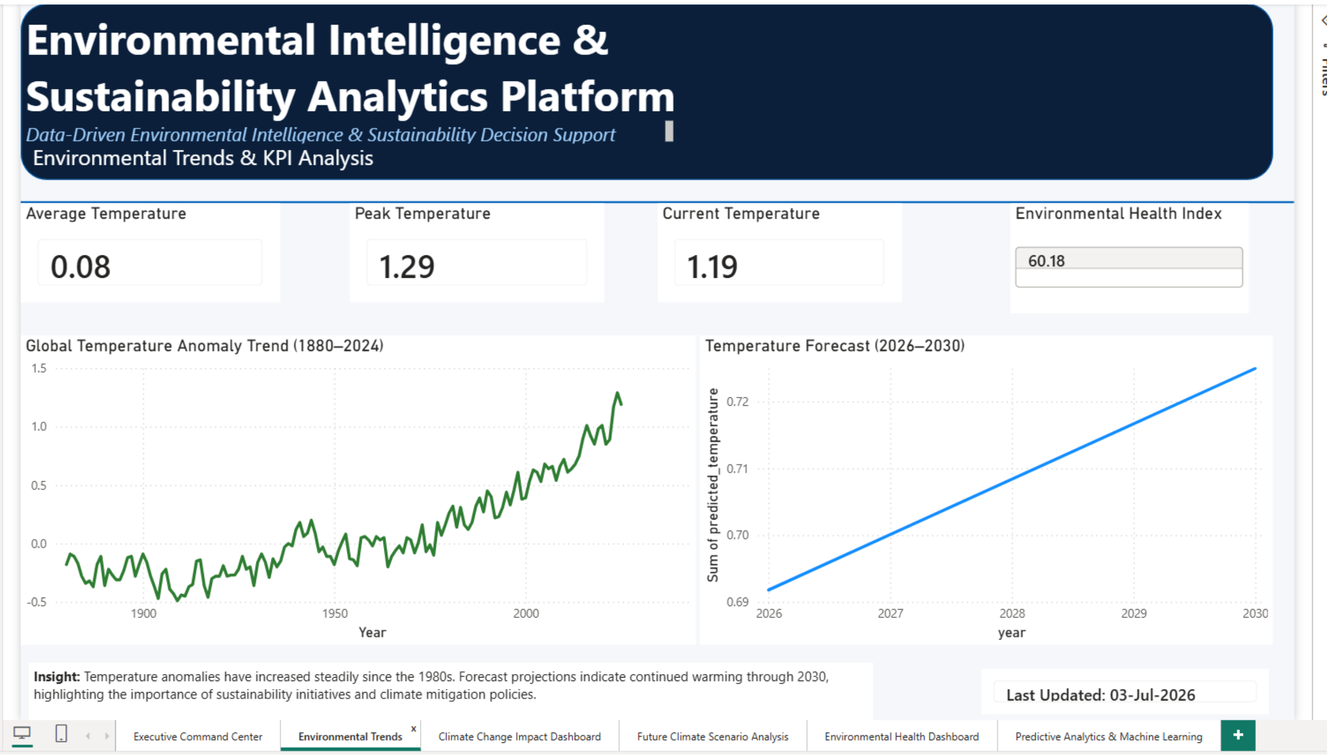
Random Forest

Selected based on superior predictive accuracy.

Executive Findings

The dashboard provides consolidated insights enabling executives to rapidly assess environmental conditions and sustainability status.

6. Environmental Trends Dashboard



Overview

This dashboard analyzes long-term climate trends and future projections.

Key Components

Temperature Trend Analysis

Visualizes global temperature anomalies from 1880–2024.

Forecast Visualization

Forecasts future temperature trends through 2030.

KPI Monitoring

Includes:

- Average Temperature
- Peak Temperature
- Current Temperature
- Environmental Health Index

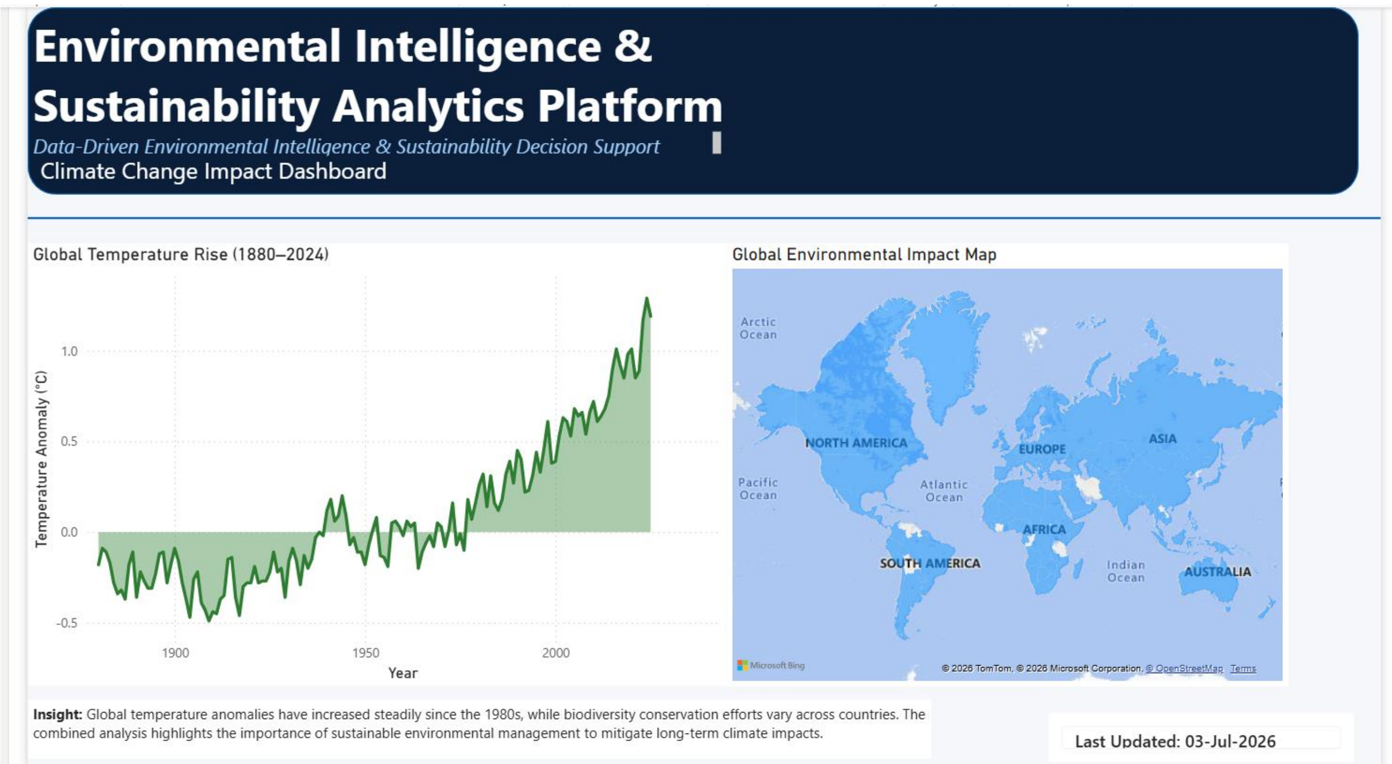
Dynamic Insights

Highlights key environmental observations and future climate implications.

Findings

The analysis indicates a persistent increase in global temperature anomalies, demonstrating continued climate change impacts.

7. Climate Change Impact Dashboard



Overview

This dashboard examines global climate impacts and geographic environmental conditions.

Key Components

Global Warming Trend

Displays historical temperature anomaly growth.

Environmental Impact Map

Provides a geographic representation of environmental impact across regions.

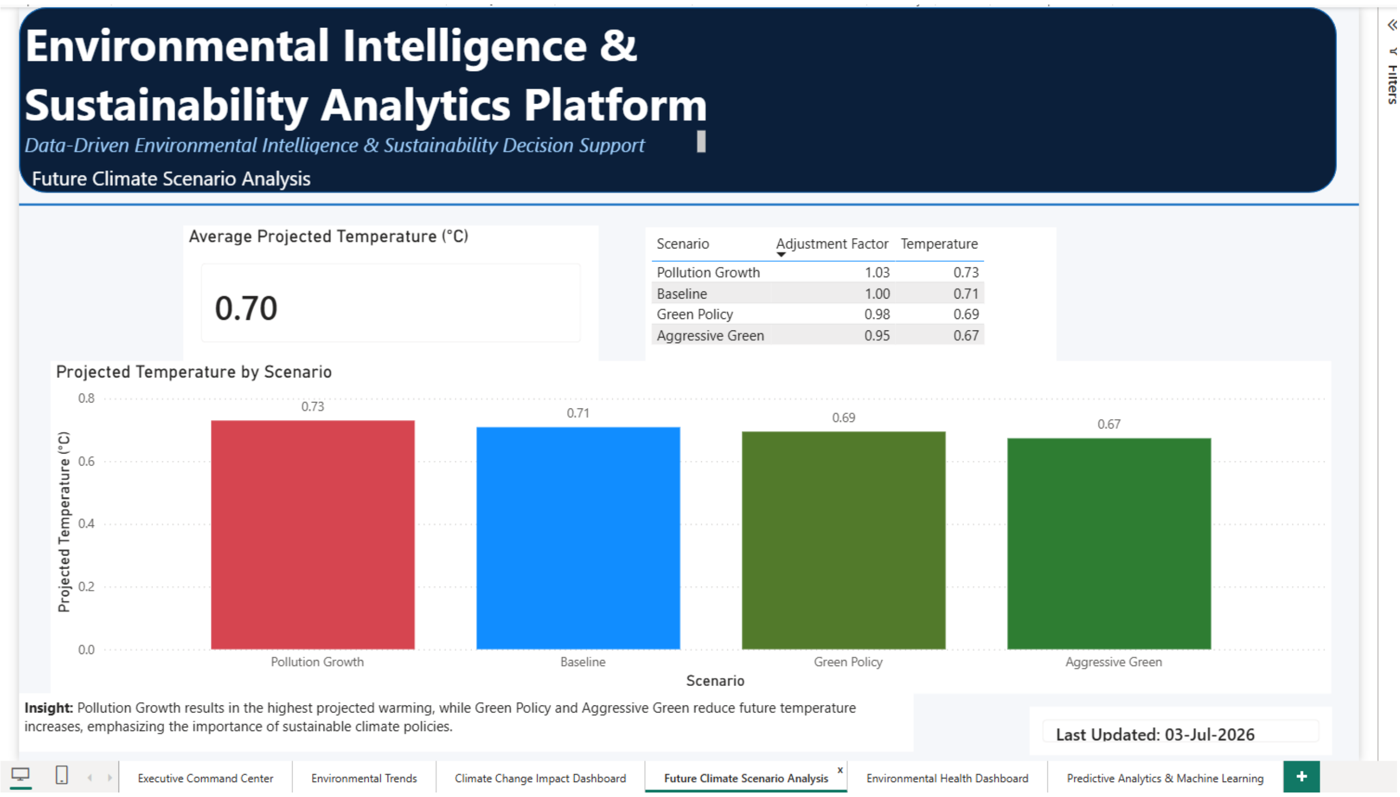
Geographic Visualization

Supports spatial analysis of environmental conditions and sustainability factors.

Findings

The dashboard demonstrates the increasing severity of climate change and emphasizes the importance of sustainable environmental management.

8. Future Climate Scenario Analysis



Overview

This dashboard evaluates possible future environmental outcomes under multiple policy scenarios.

Scenarios Evaluated

Baseline

Represents the current environmental trajectory.

Green Policy

Assumes moderate sustainability initiatives.

Aggressive Green

Assumes strong environmental intervention and conservation programs.

Pollution Growth

Represents worsening environmental conditions and increased pollution levels.

Scenario Comparison

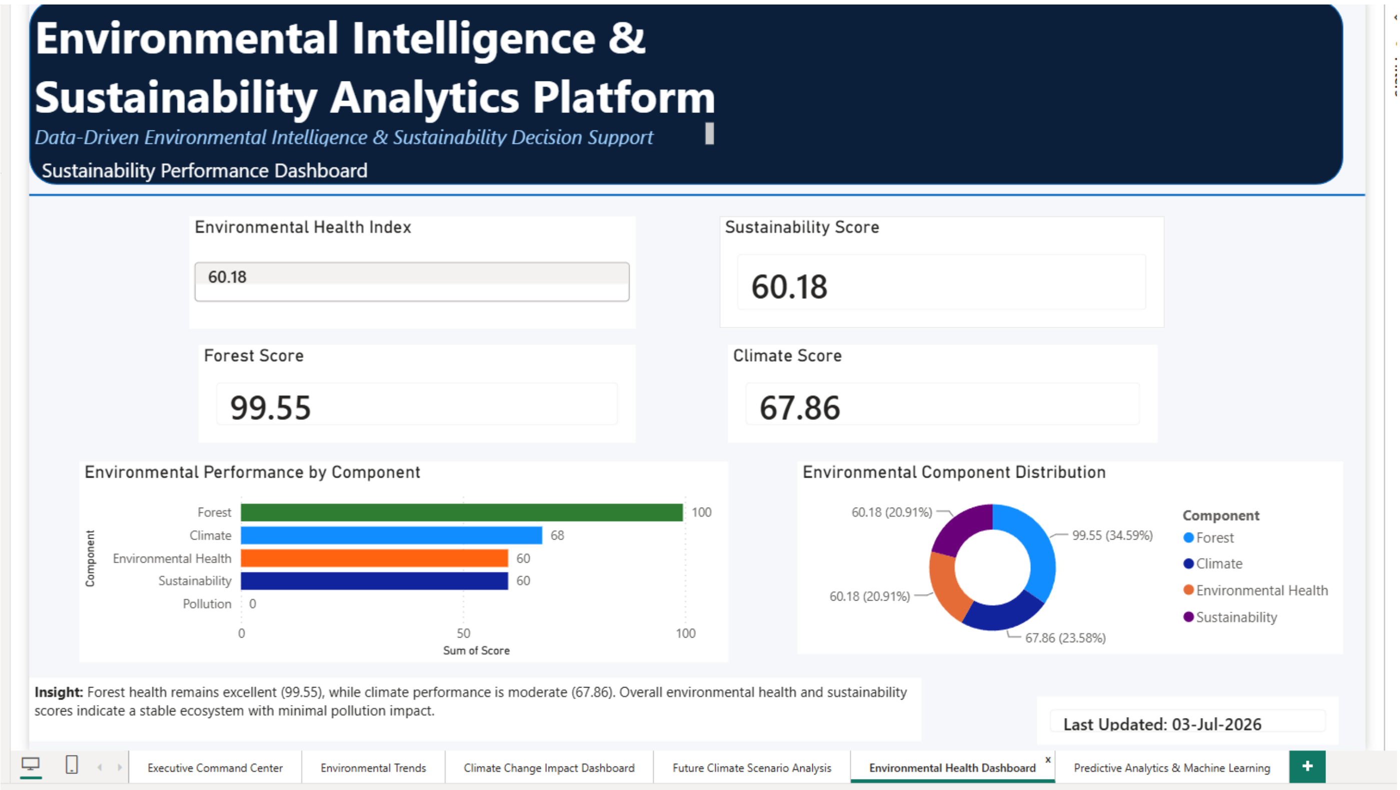
Projected temperatures indicate:

Scenario	Projected Temperature
Pollution Growth	0.73°C
Baseline	0.71°C
Green Policy	0.69°C
Aggressive Green	0.67°C

Findings

Aggressive Green generates the most favourable environmental outcome, while Pollution Growth produces the highest projected warming.

9. Environmental Health Dashboard



Overview

This dashboard evaluates environmental sustainability performance across key indicators.

Key Metrics

Environmental Health Index

60.18

Sustainability Score

60.18

Forest Score

99.55

Climate Score

67.86

Component Distribution

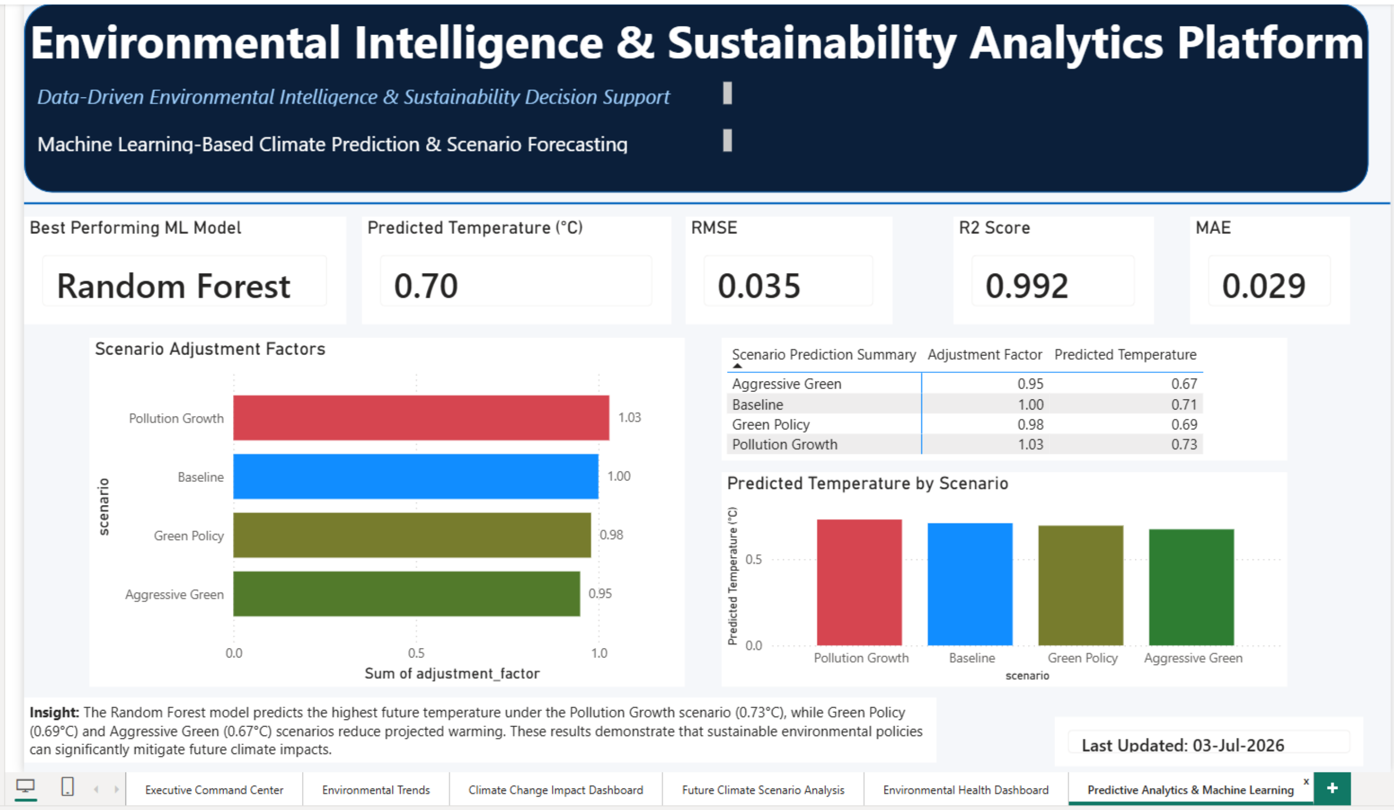
Environmental performance is analyzed through:

- Forest Sustainability
- Climate Performance
- Environmental Health
- Sustainability Score
- Pollution Indicators

Findings

Forest sustainability remains the strongest environmental component, while overall ecosystem performance remains stable and sustainable.

<u>10. Predictive Analytics & Machine Learning Dashboard</u>



Overview

This dashboard presents machine learning forecasting and evaluation results.

Best Model

Random Forest Regressor

Model Evaluation Metrics

Metric	Result
R^2 Score	0.992
RMSE	0.035
MAE	0.029

Scenario Prediction Analysis

Machine learning forecasts predict environmental outcomes under different future scenarios.

Machine Learning Validation

The exceptionally high R^2 score confirms excellent predictive capability and model reliability.

Findings

Random Forest provided the highest forecasting accuracy and was selected as the production forecasting model.

11. Key Findings

The Week 6 dashboards generated several important insights:

- Global temperature anomalies have increased consistently over time.
- Climate change impacts continue to intensify.
- Pollution Growth produces the highest projected temperature outcomes.
- Aggressive Green policies result in the lowest projected warming levels.
- Forest sustainability remains the strongest environmental component.
- Environmental Health Index indicates moderate ecosystem health.
- Sustainability performance is currently classified as good.
- Random Forest achieved exceptional predictive performance with an R^2 score of approximately 0.99.
- Geographic visualization demonstrates the global nature of environmental challenges.
- Data-driven sustainability initiatives can significantly reduce future environmental risks.

12. Deliverables Completed

The following Week 6 deliverables were successfully completed:

☑ Interactive Power BI Dashboard Platform

☑ Executive Command Center

☑ Environmental Trends Dashboard

☑ Climate Change Impact Dashboard

☑ Future Scenario Analysis Dashboard

☑ Environmental Health Dashboard

☑ Predictive Analytics & Machine Learning Dashboard

☑ Geographic Map Visualization

☑ Executive KPI Framework

☑ Forecast Visualization

☑ Sustainability Reporting

☑ Dashboard Architecture Diagram

☑ Decision-Support Insights

13. Challenges Encountered

Several challenges were addressed during development:

Dashboard Optimization

Balancing visual performance and dashboard responsiveness.

Multiple Dataset Integration

Combining climate, pollution, and forest sustainability datasets into a unified reporting framework.

Geographic Visualization

Creating meaningful map-based environmental analysis.

Machine Learning Presentation

Translating technical model outputs into business-friendly insights.

Interactive Dashboard Design

Ensuring dashboards remain informative while maintaining simplicity and usability.

14. Conclusion

Week 6 successfully transformed the analytical outputs developed during Week 5 into a fully interactive Environmental Intelligence & Sustainability Analytics Platform.

The completed solution integrates environmental monitoring, sustainability assessment, machine learning forecasting, scenario analysis, geographic visualization, and executive reporting within a unified Power BI environment.

The platform enables environmental stakeholders, policymakers, and sustainability managers to make informed, data-driven decisions through actionable insights and predictive analytics.

15. Week 6 Achievements

During Week 6, the project successfully transformed engineered datasets, predictive models, and scenario analysis into an integrated Business Intelligence platform. The completed dashboards enable interactive exploration of climate trends, environmental health indicators, sustainability metrics, geographic analysis, and predictive forecasting. The solution demonstrates the practical application of data engineering, machine learning, and visualization techniques for executive decision support.

16. Future Scope

Potential future enhancements include:

- Real-time IoT environmental sensor integration
- Cloud-based deployment and automated refresh
- Live environmental monitoring dashboards
- AI-powered sustainability recommendations
- Automated predictive alert systems
- Advanced climate risk simulations

- Enterprise-scale sustainability governance reporting

Week 6 Outcome

Week 6 Deliverable Status: Completed Successfully

Deliverables Achieved: 100%

Output Produced: Executive-Ready Environmental Intelligence & Sustainability Analytics Platform.